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Database issue

DDInter 2.0: an enhanced drug interaction resource with expanded data coverage, new interaction types, and improved user interface

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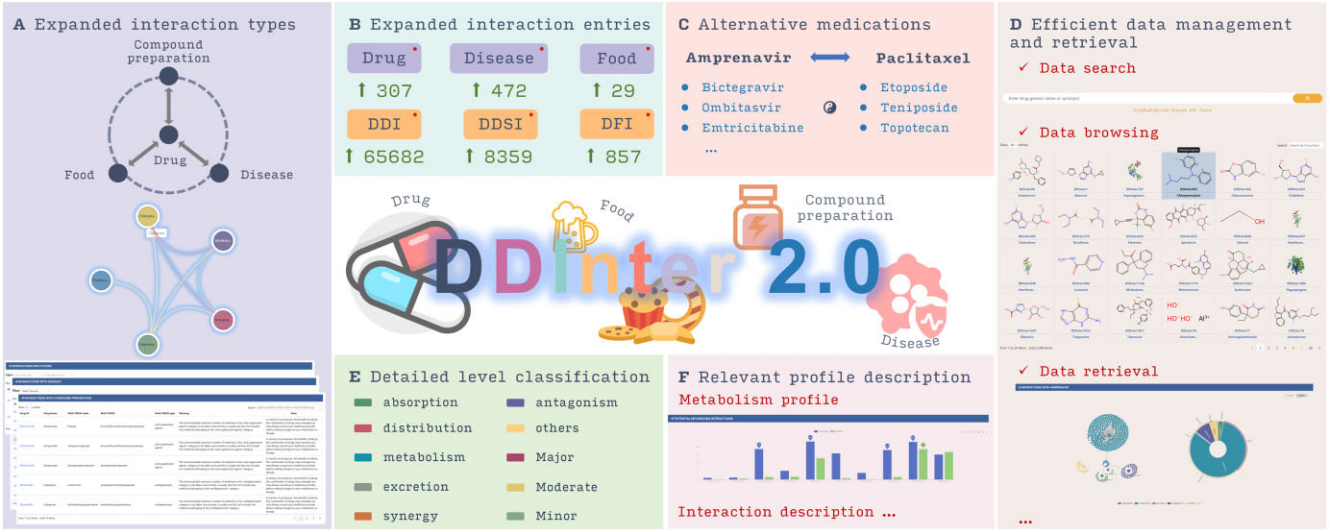
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Abstract

Drug interactions pose significant challenges in clinical practice, potentially leading to adverse drug reactions, reduced efficacy, and even life-threatening consequences. As polypharmacy becomes increasingly common, the risk of harmful drug interactions rises, underscoring the need for comprehensive and user-friendly drug interaction resources to ensure patient safety. To address these concerns and support healthcare professionals in optimizing drug therapy, we present DDInter 2.0, a significantly expanded and enhanced update to our drug interaction database. This new version incorporates additional interaction types, including drug-food interactions (DFIs), drug-disease interactions (DDSIs), and therapeutic duplications, providing a more complete resource for clinical decision-making. The updated database covers 2310 drugs, with 302 516 drug–drug interaction (DDI) records accompanied by 8398 distinct, high-quality mechanism descriptions and management recommendations. DDInter 2.0 also includes 857 DFIs, 8359 DDSIs and 6033 therapeutic duplication records, each supplemented with detailed information and guidance. Furthermore, the enhanced user interface and advanced filtering options in this second release facilitate easy access to and analysis of the comprehensive drug interaction data. By providing healthcare professionals and researchers with a more complete and user-friendly resource, DDInter 2.0 aims to support clinical decision-making and ultimately improve patient outcomes. DDInter 2.0 is freely accessible at https://ddinter2.scbdd.com.

Graphical abstract



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Introduction

Drug–drug interactions (DDIs) pose a significant challenge in clinical practice, as they can result in adverse drug reactions, diminished therapeutic efficacy, and in some cases, life-threatening outcomes (1). The likelihood of harmful DDIs escalates as the number of drugs taken by a patient increases (2), especially for those accompanied by chronic diseases such as cancer, diabetes, and cardiovascular disorders (3). Studies have suggested that DDIs may be responsible for up to 30% of all reported adverse drug events, contributing to a rise in hospitalization rates and emergency department visits (4,5). To enhance drug safety and optimize therapeutic outcomes, it is crucial to thoroughly understand the pharmacological properties of each medication before initiating combination therapy, thereby minimizing the risk of adverse drug reactions while maximizing the desired therapeutic effects (6).

To assist clinicians in screening for dangerous drug combinations and improve clinical decision-making, we previously developed DDInter, a curated DDI database with comprehensive data, practical medication guidance, intuitive function interface, and powerful visualization. DDInter aimed to provide healthcare professionals with a reliable and user-friendly resource for optimizing drug therapy and ensuring patient safety. The first version of DDInter contained about 0.24M DDI associations connecting 1833 approved drugs (1972 entities), with each drug annotated with basic chemical and pharmacological information and its interaction network (7).

Despite the extensive coverage of DDIs in DDInter, other types of drug interactions, such as drug-food interactions (DFIs), drug-disease interactions (DDSIs), and therapeutic duplications, were not fully addressed. Although DrugBank (8) provides some DFI information, it is limited in scope. Moreover, other freely available DDI databases, such as SuperCYPsPred (9) and SuperDRUG2 (10), rarely include DFIs, focusing primarily on DDIs. Commercial databases like DIDB provide detailed DDI information and some DFI and DDSI data, as well as interaction mechanisms and medication recommendations. However, their high subscription fees restrict access for healthcare professionals in primary care settings (11). The limited and incomplete annotations for these types of interactions, coupled with restricted access to commercial databases, underscore the need for their inclusion in comprehensive drug interaction resources.

DFIs and DDSIs are increasingly recognized as crucial factors in clinical decision-making and patient safety (5,12–14). With the growing consumption of food supplements, functional foods, and herbal medicinal products, the risk of interactions between bioactive compounds and pharmaceutical products has raised concerns (15,16). DDSIs are also prevalent, as evidenced by a recent study indicating that 13.9% of all prescriptions filled in community pharmacies triggered a drug therapy alert related to a DDSI (17). These interactions can lead to adverse drug reactions, reduced efficacy, and increased healthcare costs. Furthermore, therapeutic duplication, which occurs when a patient is prescribed two or more drugs from the same therapeutic class, can result in an increased risk of adverse effects, overdose, and unnecessary healthcare expenditure (18).

To address these limitations and provide a more comprehensive resource for clinical decision support, we have developed DDInter 2.0. In this updated version, the database now covers 2310 drugs (2122 distinct drugs) and contains 302 516 DDI records, accompanied by 8398 distinct and high-quality mechanism descriptions and management recommendations. Additionally, DDInter 2.0 includes 857 DFIs involving 29 foods, with 430 high-quality mechanism descriptions and management recommendations, 8359 DDSIs involving 472 diseases with 3300 detailed interaction information and management recommendations, and 6033 therapeutic duplication records involving 317 combination drugs and 96 pharmacological classes, each accompanied by a specific warning and note. To ensure the accuracy and clinical relevance of the information, we have excluded any newly annotated interactions marked as 'unknown' in this new version. A detailed comparison of the data between the new and old versions can be found in Table 1.

Materials and methods

Data collection

To comprehensively collect DDI information, this study consistently retrieved literature reporting the influence of one drug on another from PubMed, similar to the approach used in DDInter, which covered research through March 2021. For DDInter 2.0, we collected publications between March 2021 and May 2023. In addition to continuing DDI literature collection, DDInter 2.0 also included reports on drug effects on food and disease, with a cutoff in May 2023. Literature unrelated to drug interactions, such as drug-target or drug-gene interactions, was excluded. In addition to the literature, we also downloaded all drug labels and guidelines approved after March 2021 from the FDA. These materials were used to extract drug interaction information and provide guidance on avoiding severe adverse events. All collected materials underwent deduplication.

Eventually, we obtained 6568 relevant literature pieces (5564 on DDIs, 845 on drug–food interactions and 6772 on drug–disease interactions). This served as the document library for subsequent annotation. It should be noted that some documents described interactions of a drug with a drug/food/disease class. We extracted interaction data using NLP from the document library. For interactions involving drug/food/disease classes, we manually processed them into individual drug pairs by referring to the original content in the document library (19). To standardize data from different sources, we constructed a drug synonym dictionary combining DrugBank and Drugs@FDA and standardized disease names using MESH.

After comparing and deduplicating new DDI data with the source database, we performed standardized deduplication on other interaction types and cleaned residual/abnormal extracted data, tracing back to the original literature when needed. For compound preparations, concurrent use of contained drugs may cause therapeutic duplication interactions. We separately classified such cases in the library.

As before, we worked with pharmacists to systematically review, grade, and annotate interaction mechanisms for the newly expanded DDI data and newly added DFI data. Each entry was independently reviewed by at least two pharmacists, with a third resolving any disagreements. Risk levels and corresponding management strategies were annotated for the newly added DDSI data. Therapeutic duplication data were provided with corresponding medication duplication warnings and notes. Notably, DDInter 2.0 did not add unknown annotations to avoid increased alert fatigue. Finally, we obtained ATC codes for the newly added drugs and updated the

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Table 1. Detailed data comparison between DDInter 1.0 and 2.0

Number of Entry	Classification	Mechanism	DDInter 1.0	DDInter 2.0	Increase
DDI	PK type	Absorption	7294	10351	42%
		Distribution	924	1597	73%
		Metabolism	42 180	57 516	36%
		Excretion	2344	3406	45%
	PD type	Synergy	110 199	134 817	22%
		Antagonism	18 970	24 723	30%
	Other Mechanism	Others	13 002	30 318	133%
	Risk Lever	Minor	9805	12522	28%
		Moderate	145 132	195 776	35%
		Major	39 480	52 943	34%
		unknown	42 417	42 417	-
	Total of DDI		236 834	302 516	28%
Drug			1972	2310	17%
DDSI				8359	New Drug Interaction Type in DDInter2.0
Disease				439	
DFI				810	
Food				29	
Therapeutic Duplication				6033	

ATC codes for existing drugs to facilitate the display of substitution options (20).

Database implementation

The DDInter2.0 online database was developed using Django 4.2.11 for the back end and Bootstrap 5.3.0 for the front end. The drug interaction data were stored in a SQLite3 database with a carefully designed schema and optimized query performance. RDKit was employed for various molecular tasks (21), whereas 3Dmol.js was utilized to showcase molecular 3D structures (22). Interactive data visualizations, including relationship graphs, sunburst charts, and bar charts, were created using ECharts 5.5 (23). DDInter 2.0 is now accessible via a dedicated server hosted on Alibaba Cloud. This server boasts 24 vCPUs, ensuring robust processing capabilities, along with 1TB of disk space to accommodate extensive data storage and 48GB of RAM to guarantee smooth performance. Prior to launch, the database underwent rigorous compatibility testing across various operating systems and modern web browsers. The DDInter 2.0 database is conveniently accessible to the public via https://ddinter2.scbdd.com, eliminating the requirement for any login credentials.

Results

Expanded DDI data coverage

Since the release of DDInter in September 2020, nearly three years passed. During this period, numerous new drugs have been approved and marketed, and scientists have confirmed more novel DDIs. To keep pace with the latest advances in the field of DDIs, we have significantly expanded the DDI data in DDInter 2.0.

Compared to the first version of DDInter, the total number of DDIs recorded in DDInter 2.0 has increased from 236 834 to 302 516, a 27.73% increase. This expansion is reflected across various interaction types, including pharmacokinetic (PK) and pharmacodynamic (PD) categories. In the PK category, interactions related to absorption increased by 41.91%, distribution by 72.84%, metabolism by 36.36%, and excretion by 45.31%. Similarly, in the PD category, synergistic interactions increased by 22.34%, antagonistic interactions by 30.33%, and interactions classified as 'other mechanisms' by 133.18%, a significantly higher increase than other types. This might have been attributed to the complexity of the interaction mechanisms of new drugs or the lack of comprehensive research on these mechanisms. On the other hand, interactions with simple mechanisms might have been thoroughly studied, and newly discovered interactions were often difficult to explain by a single mechanism.

DDInter 2.0 employed an enhanced risk level classification system, with mild, moderate, and severe risk levels expanded by 27.71%, 34.90% and 34.10%, respectively. The similar increase across these three risk levels indicates a balanced distribution of newly discovered DDIs across different severity levels. This phenomenon may reflect the comprehensiveness of DDI research and the potential for new drugs to exhibit interactions at various risk levels. Furthermore, the number of drugs covered in the database has increased from 1972 to 2310, a growth of 17.14%. This expansion ensured that DDInter can provide relevant and comprehensive references for a broader range of drugs. In addition, DDInter 2.0 has significantly increased the number of distinct and high-quality mechanism descriptions and management recommendations from 5560 entries in DDInter 1.0 to 8398 entries, an increase of 51.04%. This substantial growth in detailed information enhances the utility of DDInter as a comprehensive reference tool, better supporting clinical decision-making and patient safety. In this version, we have excluded new interaction data with unknown mechanisms, which helps reduce alert fatigue.

Integration of new interaction types

DDInter 2.0 incorporates three new types of drug interaction data: 8359 drug–disease interactions (DDSI), 857 drug–food interactions (DFI) and 6033 therapeutic duplication interactions. These additions significantly enhance the comprehensiveness and utility of our drug interaction knowledge base by expanding the coverage of drug interactions and providing more clinically relevant information.

Analysis of the DDSI data revealed that kidney diseases, liver diseases, and diabetes mellitus were the most common diseases associated with drug-disease interactions (Figure 1). This finding underscores the importance of considering

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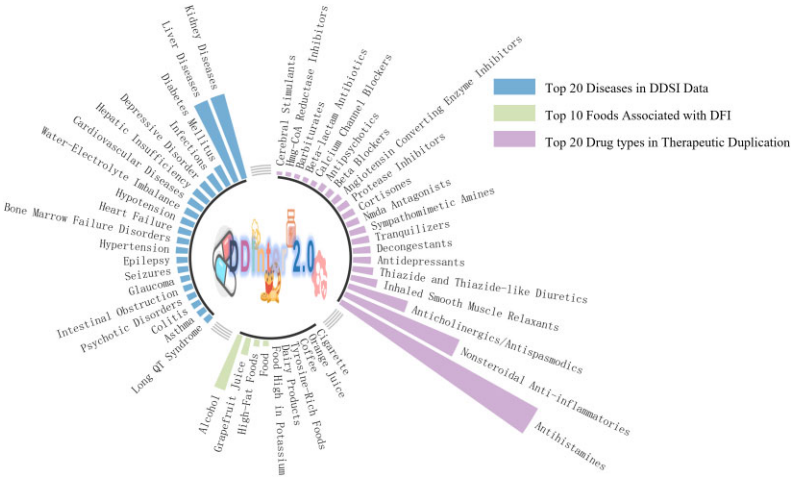


Figure 1. Most prevalent diseases, foods, and drug types in DDInter 2.0 interactions. DDSI: Drug—Disease Interactions—interactions between drugs and specific diseases/conditions that can impact drug safety or efficacy. DFI: Drug–Food Interactions—interactions between drugs and food items/dietary components that can alter drug pharmacokinetics or pharmacodynamics. Therapeutic Duplication—concurrent use of multiple drugs from the same therapeutic class, increasing adverse effect and overdose risk.

patient comorbidities when prescribing medications to avoid potential adverse events. In clinical practice, patients with these conditions require extra caution when receiving pharmacotherapy, as their ability to metabolize or eliminate drugs may differ significantly from healthy individuals. For instance, healthcare providers should be vigilant when prescribing medications with known hepatotoxicity or nephrotoxicity to patients with liver or kidney diseases, as impairment of these organs may lead to drug accumulation in the body, increasing the risk of adverse reactions (24,25). Regular monitoring of liver and kidney function tests and evaluation of patient clinical symptoms are crucial in these cases. DDInter 2.0 provides 3300 detailed interaction information and management recommendations, covering a total of 472 diseases, which can greatly assist healthcare providers in making informed decisions and optimizing patient care. Healthcare providers should exercise caution when prescribing these medications to patients with the aforementioned comorbidities and closely monitor them for any signs of adverse reactions.

The DFI data provided valuable insights into the role of food and beverages in drug interactions. Alcohol and grapefruit juice were the most common food items associated with DFI (Figure 1), underscoring the need for patient education on the potential risks of consuming these items while taking certain medications. Alcohol consumption can alter the PK and PD of various drugs, leading to reduced efficacy or increased toxicity. Similarly, grapefruit juice is known to inhibit cytochrome P450 enzymes, which can result in elevated drug levels and a higher risk of adverse events, particularly for medications like statins, calcium channel blockers, and certain antidepressants (26). By integrating DFI data into DDInter 2.0, we aim to promote safer medication use and reduce the incidence of preventable adverse events through better-informed clinical decision-making and patient counseling.

Notably, DDInter 2.0 includes 430 high-quality mechanism descriptions and management recommendations for DFI, covering various aspects such as absorption (189), metabolism (210), synergy (390), antagonism (15), excretion (1) and others (53). This comprehensive coverage enables healthcare providers to understand the underlying mechanisms of DFI and take appropriate measures to mitigate their impact on patient care. Therapeutic duplication statistics provide valuable insights into the prevalence of potentially inappropriate medication use. Our analysis revealed that antihistamines, nonsteroidal anti-inflammatory drugs (NSAIDs), and anticholinergics/antispasmodics were the most common drug classes involved in therapeutic duplication (Figure 1). This type of duplication can occur for various reasons, such as prescriptions from multiple healthcare providers or inadequate medication reconciliation (18). For example, a patient may unknowingly take a cold remedy and a sleep aid, both containing diphenhydramine, or receive prescriptions for a sleep aid and an antianxiety drug with similar sedative properties from different doctors. Such duplication can result in adverse events, such as excessive sedation. To prevent and address therapeutic duplication, it is crucial to implement strategies that enhance communication among healthcare providers, educate patients on proper medication use, and ensure regular medication reviews (27). By addressing therapeutic duplication, clinicians can streamline patient medication regimens, reduce the risk of adverse events, and improve overall patient outcomes. DDInter 2.0 includes data on 317 combination drugs and 96 pharmacological classes, each accompanied by a specific warning and note. This information can help healthcare providers identify and prevent potential therapeutic duplications, leading to safer and more effective medication use.

Enhanced database functions and visualization

We have successfully integrated ADMET 3.0 (28), a cutting-edge tool capable of predicting crucial properties like Cytochrome P450 (CYP) interactions, into DDInter 2.0 (Figure 2B). This integration empowers DDInter 2.0 to reevaluate molecular interaction properties, resulting in more precise and comprehensive predictive outcomes. By leveraging this update, not only does the reliability and utility of the database receive a significant boost, but users also benefit from a more robust data foundation tailored specifically for drug design and toxicology research.

To elevate user experience and streamline data analysis, we've implemented substantial optimizations to the interface

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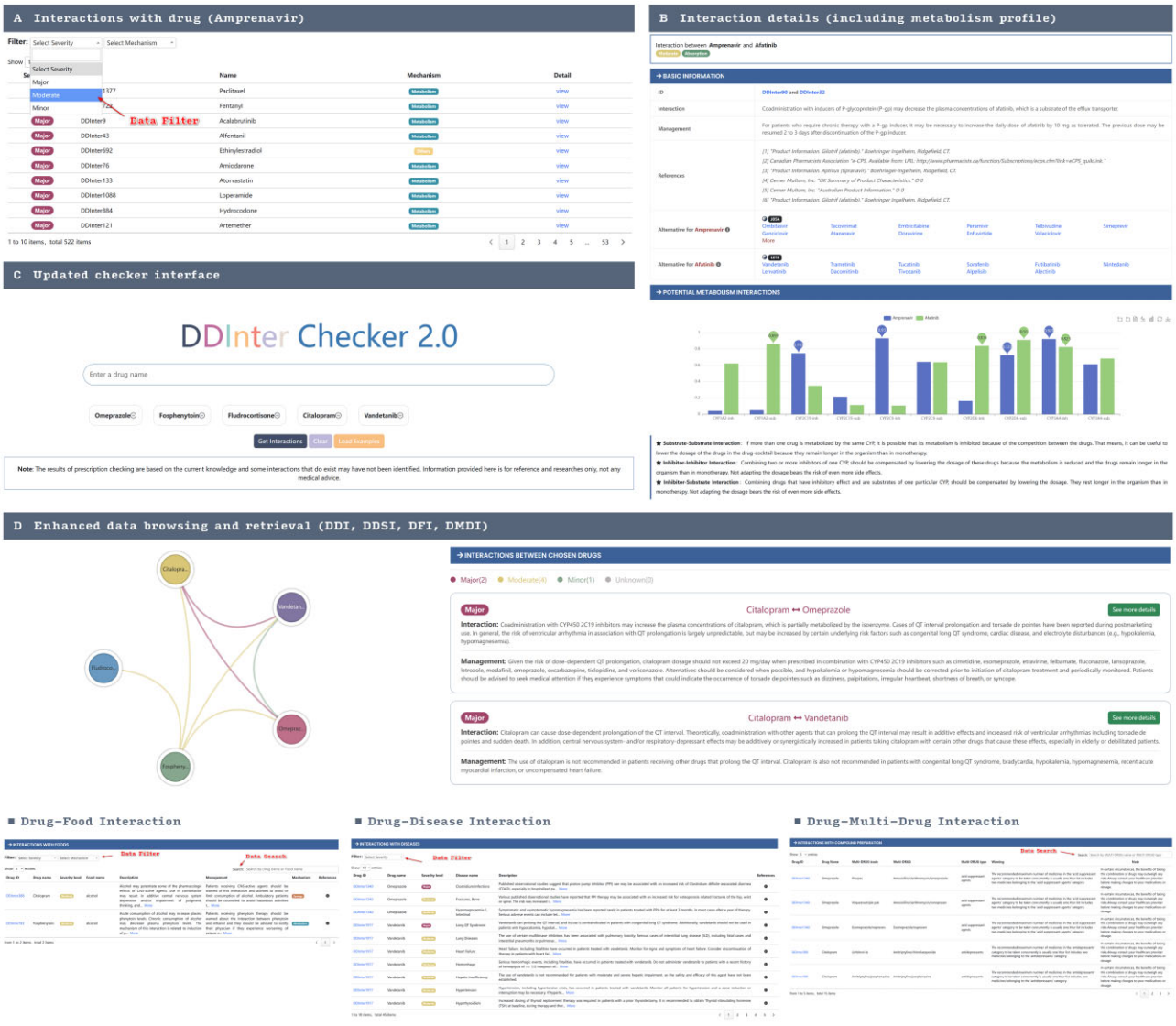


Figure 2. The web interface of DDInter 2.0. (**A**) A list of drugs interaction with the query drug (Amprenavir, DDInter90). (**B**) The page of a specific drug-drug pair shows extensive information on interactions, including alternative drug recommendations. (**C**) The interaction checker module helps physicians screen for risks in prescriptions. (**D**) Results of interaction checker (example) and enhanced interaction data management interface.

of DDInter 2.0, coupled with enhancements to the filtering functionality. As shown in Figure 2A, the Filter component offers mechanism-based and risk-based filtering options, allowing users to quickly and accurately screen the information they need. The redesigned interface is characterized by enhanced intuitiveness and user-friendliness, facilitating effortless data access and comprehension. Furthermore, we've bolstered the filtering and searching capabilities by introducing additional criteria and options, enabling users to swiftly locate desired information and conduct thorough data analysis (Figure 2). These improvements are geared towards empowering users to make more effective decisions based on comprehensive insights.

It's essential to highlight that DDInter's entities and interface undergo continual minor improvements and revisions. Minor corrections are made without formal announcements. Significant changes, defined as those exceeding 10% in components, are duly noted in descriptions alongside modified update dates.

Conclusion

DDInter 2.0 represents a significant advancement in the field of drug interaction management, providing healthcare professionals with a comprehensive, user-friendly, and reliable resource to support clinical decision-making and promote patient safety. This updated version features an expanded DDI dataset, with 302 516 DDI records covering 2310 drugs, and includes 8398 distinct, high-quality mechanism descriptions and management recommendations.

In addition to the enhanced DDI data, DDInter 2.0 integrates three new types of drug interaction data: DDSI, DFI and therapeutic duplication interactions. The database now contains 8359 DDSI records with 3300 detailed interaction mechanisms and management recommendations, covering 472 diseases. It also includes 857 DFI records with 430 high-quality mechanism descriptions and management recommendations, as well as 6033 therapeutic duplication records involving 317 combination drugs and 96 pharmacological classes, each accompanied by a specific warning and note.

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Moving forward, the DDInter team is dedicated to continuously updating and improving the database, ensuring its position as a trusted and essential resource for healthcare providers. Future updates will focus on maintaining the high quality of annotations and practical medication guidance that have become synonymous with DDInter while also exploring the inclusion of additional interaction types, such as drug—herb interactions, to further enhance the database's comprehensiveness and relevance to a broader range of healthcare settings.

Efforts will be directed towards improving the user experience by optimizing visualization and interaction checker functions, making it easier for healthcare professionals to access and utilize the wealth of information contained within DDInter. By actively seeking and incorporating user feedback, the database will continue to evolve and adapt to the changing needs of its users, maintaining its position at the forefront of drug interaction management.

In conclusion, DDInter 2.0 represents a substantial step forward in providing healthcare professionals with a powerful tool to manage the complexities of drug interactions. Through its expanded data coverage, integration of new interaction types, and enhanced functionality, this updated version demonstrates a strong commitment to comprehensive data coverage, practical medication guidance, user-friendly design, and powerful results visualization. As the landscape of drug interaction research continues to evolve, DDInter will remain steadfast in its mission to support clinical decision-making and promote patient safety in the era of increasing polypharmacy.

Data availability

DDInter 2.0 is freely accessible at https://ddinter2.scbdd.com.

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Conflict of interest statement

None declared.

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